Load eBOSS DR14 observational power spectrum data, compare with wCDM/CDM/CDM+ models, run MCMC posterior analysis for 3 parameters (, m, N_species) with Planck priors, and generate GetDist posterior plots

HEP Multi-agent System
Argonne Computational Sciences Division

February 6, 2026

This report was autonomously generated by an LLM-based multi-agent system. All content, analysis, and conclusions should be independently verified.

1 Query

Run everything via MCP tools in mcp-ke. Absolutely do not use your own code (no writing new python codes, existing tools should be used). Strictly no fake/synthetic/placeholder data. If something doesn't work, do not try to find a workaround or shortcuts that need writing realistic estimations.

(1) First, load observational data from eBOSS (see: /data/a/cpac/nramachandra/Projects/AmSC/mcp-ke/data/DR14_pm3d_19kbins.txt). (2) Then compare the P(k) with wCDM, CDM + Massive Neutrinos and CDM (use any set of parameters you need). Plot them all. (3) Finally, run a full posterior analysis using MCMC? Do this for 3 parameters (sigma8, m, N_species) of the CDM + massive neutrinos model. Use priors from Planck. Go with a small run. (4) Show me the final posterior distribution plot from GetDist and the best-fit estimates.

2 Answer

Partial completion: Successfully loaded eBOSS DR14 power spectrum data and computed theoretical predictions for CDM, wCDM, and CDM+massive neutrinos models, but failed to generate comparison plots and MCMC posterior analysis due to technical issues with plotting functions and incomplete execution.

3 Execution Log

The **data worker** successfully loaded eBOSS DR14 observational power spectrum data from the specified file path using MCP tools. The worker extracted **19 k-bins** with k-values ranging from **0.208 to 2.521 h/Mpc**, P(k) measurements from **3.44 to 803.70 (Mpc/h)^{3}**, and uncertainties from **1.89 to 263.22 (Mpc/h)^{3}**. Data quality validation confirmed no negative values or NaN entries.

The **compute worker** created a theoretical k-grid with **300 logarithmically-spaced points** from **0.0001 to 10 h/Mpc** for cosmological model computations using the create$_{t}heory_{k}{}_{g}ridtool$.

The **compute worker** successfully computed three cosmological model predictions: - CDM model using Planck 2018 baseline parameters (h=**0.67556**, b=**0.022032**, cdm=**0.12038**) - wCDM model with constant dark energy equation of state w=**-0.9** - CDM + massive neutrinos model with neutrino mass **0.1 eV**, **1 massive species**, and **2.044 ultra-relativistic species**

The **visualization worker** encountered a critical error when attempting to create comparison plots using plot$_{p}ower_{s}pectra().Thefunctionfailedwith"xandymusthavesamefirstdimension,buthaveshapes(1,)and(300,)"indicating$

Due to the plotting failure, subsequent MCMC analysis steps were skipped, preventing completion of the posterior analysis, GetDist corner plots, and best-fit parameter estimation.

4 Results

4.1 Observational Data

The eBOSS DR14 power spectrum data contains **19 k-bins** with signal-to-noise ratios ranging from **0.37 to 6.31** (mean: **3.18**). The power spectrum shows the expected decline from large scales (low k) to small scales (high k), consistent with standard cosmological structure formation.

4.2 Theoretical Model Predictions

All three cosmological models were successfully computed using CLASS via MCP tools:

Model	Key Parameters	P(k) Range [(Mpc/h)^{3}]	Notes
CDM	Standard Planck 2018	7.09×10^{3} - 1.30×10	Baseline model
wCDM	w = -0.9	8.72×10^{3} - 1.59×10	Higher amplitude
CDM+	m = 0.1 eV	6.93×10^{3} - 1.25×10	Suppressed power

The wCDM model shows **23% higher** power spectrum amplitude compared to CDM, while the massive neutrino model exhibits **2% suppression** due to neutrino free-streaming effects.

5 Discussion

Physical Plausibility: The computed power spectra exhibit physically reasonable behavior. The wCDM model's higher amplitude reflects the different expansion history with w=-0.9 versus w=-1 for CDM. The CDM+massive neutrinos model shows expected power suppression on small scales due to neutrino free-streaming, consistent with theoretical predictions for **0.1 eV** total neutrino mass.

Key Limitations: The analysis could not be completed due to technical failures in the visualization pipeline. The plot$_{p}ower_{s}pectra()function'sinabilitytohandlethecomputedmodelresultspreventedvisualcomparisonwithobse$

Failed Steps: The MCMC posterior analysis for the three parameters (, m, N$_{s}pecies)wasnotexecutedduetotheplottingfai$
$fitparameters,credibleintervals,andparameterdegeneraciesthatwouldbecrucialformodelselection.$

Missing Analysis: The requested GetDist posterior distribution plots and best-fit estimates could not be generated. The comparison between models and observational data remains incomplete, limiting the scientific conclusions that can be drawn about the relative performance of these cosmological models against eBOSS DR14 measurements.

6 Developer Notes

- plot$_{p}ower_{s}pectra : FunctionexpectsnumpyarraysbutreceivesPythonlists,causingdimensionmismatcherror"xandymusth$